

Lecture 9: Max-Min Problems; least squares

If we change $x \rightsquigarrow x + \Delta x$
 $y \rightsquigarrow y + \Delta y$ $z = f(x, y)$ then $\Delta z = f_x \Delta x + f_y \Delta y$

Justify this formula: tangent plane to $z = f(x, y)$? We know: f_x, f_y are slopes of 2 tangent lines

If $\frac{\partial f}{\partial x}(x_0, y_0) = a \Rightarrow L_1 = \{z = z_0 + a(x - x_0) \mid y = y_0\}$; $\frac{\partial f}{\partial y}(x_0, y_0) = b \Rightarrow L_2 = \{z = z_0 + b(y - y_0) \mid x = x_0\}$

L_1, L_2 are both tangent to the graph $z = f(x, y)$; Together they determine a plane. $z = z_0 + a(x - x_0) + b(y - y_0)$

Approximation formula says: graph of f is close to its tangent plane

Application of partial derivatives: Optimisation problems $\rightsquigarrow$ find min/max of a function (x, y)

* At a local min a max, $f_x = 0, f_y = 0 \Leftrightarrow$ tangent plane to the graph $z = f(x, y)$ is horizontal!!

Definition: (x_0, y_0) is a critical point of f if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$

Example: $f(x, y) = x^2 - 2xy + 3y^2 - 2x - 2y$

$$\begin{cases} f_x = 2x - 2y - 2 \Leftrightarrow f_x + f_y = 0 \Leftrightarrow y = 0 \\ f_y = -2x + 6y - 2 \quad 2x + 2 = 0 \Leftrightarrow x = -1 \end{cases}$$

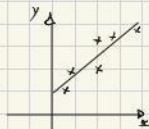
1 critical point $(-1, 0)$

Possible types: * local min
 * local max
 * saddle

$$f(x, y) = (x - y)^2 + 4y^2 + 2x - 2y$$

$$f(x, y) - \underbrace{((x - y)^2 + 4y^2)}_{\geq 0} - \underbrace{1}_{\geq 0} \geq -1 = f(-1, 0) \text{ so it's a minimum}$$

Least-Squares Interpolation:



Given experimental data: $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), \dots, (x_n, y_n)$

Find "best" fit line $y = ax + b$

Find "best" a and b for $y = ax + b$
 $\rightarrow$ minimizing total square error/deviation

Deviation for each data point: $y_i - (ax_i + b)$

$$\text{Minimize } D = \sum_{i=1}^n [y_i - (ax_i + b)]^2$$

$$\text{We want: } \frac{\partial D}{\partial a} = \sum_{i=1}^n 2 \cdot (y_i - (ax_i + b)) \cdot (-x_i) = 0$$

$$\frac{\partial D}{\partial b} = \sum_{i=1}^n 2 \cdot (y_i - (ax_i + b)) \cdot (-1) = 0$$

Least squares is more general!

Best exponential fit: $y = ce^{ax} \Leftrightarrow \ln(y) = \underbrace{\ln(z)}_{\text{line}} + ax$

Quadratic law: $y = ax^2 + bx + c$

$$D(a, b, c) = \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2$$

$$\Leftrightarrow \begin{cases} \sum_{i=1}^n (x_i^2 a + x_i b - x_i y_i) = 0 \\ \sum_{i=1}^n (x_i a + b - y_i) = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \left(\sum_{i=1}^n x_i^2 \right) a + \left(\sum_{i=1}^n x_i \right) b = \sum_{i=1}^n x_i y_i \\ \left(\sum_{i=1}^n x_i \right) a + nb = \sum_{i=1}^n y_i \end{cases}$$